

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	G. Uday Siva Sai	Reg. No.:	192425259
Section / Batch:	2024	Date:	12/07/26

Code of Conduct

I, G. Uday Siva Sai (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Annexure A – Application Based Questions

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

- Extract the candidate's name.
- Identify email addresses.
- Extract mobile numbers.
- Detect technical skills (Python, Java, SQL, Machine Learning, NLP).
- Extract years of experience.
- Generate a structured summary of the candidate's profile.
- Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

- Search products using an exact keyword.
- Perform prefix-based searches.
- Perform suffix-based searches.
- Search using partial keywords.
- Perform case-insensitive searches.
- Display all matching product names.
- Generate a report showing the total number of matching products for each search.

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

- Validate the student register number.
- Validate the institutional email address.
- Validate the course code.
- Validate the semester information.
- Validate the student's mobile number.
- Display appropriate validation messages for each field.
- Generate a final registration status report indicating whether the registration is successful.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

COI-AT1

G. Uday Siva Sai

192425259

DSA0301 - NLP

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Resume information Regular Expressions

Extraction System using

Problem Statement :- A Recruitment Company receives thousands of Resumes daily. HR wants an automated system that extracts information and identifies eligible candidates based on experience and python skills.

Algorithm :-

- 1) import the re module
- 2) store resume details in a text file
- 3) Extract the candidates name using Regexp
- 4) Extract the email address
- 5) Detect technical skills
- 6) Display eligibility status

Python Program :-

```
import re

resume = " "

enter Name, email, Phn, Skills / Experiences

name = re.search(r"Name : (.*)", resume)
```

```

email = re.search ( r' s+ @ l s+ ", resume)
phn = re.search ( r" / bld {10} \b ", resume)
skills = re.findall ( r" Python | Java | SQL, ML
                                experience )
Print ( " Candidate name ", name.group(1))
Print ( " Email ", email.group(1))
Print ( " phone ", Phone.group(1))
Print ( " experience : ", Experience.group(1).)
if int ( Experience group(1)) >= 2 and " Python " in
skills :
    Print ( " Eligible for shortlist ")
else :
    Print ( " Not eligible ")

```

Structured output :-

```

Name      : Uday
Email     : uday@gmail.com
ph        : 999921345
Skills    : Py, Java, SQL, ML
Experience : 4 years
Status    : Eligible

```


②

Product Search System using Regular Expression

Problem Statement :-

- * Design a product search system support
 - * Exact Search
 - * Prefix Search
 - * Suffix Search
 - * Partial Search
 - * Case insensitive Search

Algorithm :-

- 1) Import Regexp module
- 2) Store product names
- 3) Accept Search Keyword
- 4) Perform
 - * Exact
 - * Prefix
 - * Suffix
- 5) Display total cost

Program :-

```
import re
product = ["products"]
keyword = input("Enter product name: ")
```

Count = 0

Print ("In matching product :")

keyword = input ("Enter Product name :")

Count = 0

Print ("In matching products :")

for product in products :

if reSearch (keyword, product, re.IGNORECASE)

Print (product)

Count + = 1

Print ("\nTotal matching products", Count)

Sample Input & output :-

Input : Apple

output : Exact match []

Suffix match []

Partial Match [Apple iPhone Apple watch "]

Case Insensitive Search

[Apple, phone, "Apple watch"]

Total matches : 2

(3)

University Registration Validation System

Problem Statement :-

- * Register Number
- * Email
- * Course Code
- * Semester
- * Mobile Number

Display Validation message

Program :-

```
import re

reg = input("Enter Reg No")
email, Course, Semester, mobile

if re.fullmatch(r"[0-9]{12}", reg):
    print("Register Number : valid")
else:
    print("Register Number : Invalid")

if re.fullmatch(r"[a-zA-Z0-9]{1}@[a-zA-Z0-9]{1}.com", email):
    print("email : valid")
else:
    print("email : Invalid")
```


if re.fullmatch(r"[A-Z]{3}[a-z]{4}", Curr).

Print ("Course code : Valid")

else : Print ("Course Code : Valid")

Sample Input :-

Reg Number : 192425259

Email : vday@fauztha.com

Course : DSA0301

Semester : I

Sample Output :-

Reg Number : Valid

Email : Valid

Course code : Valid

Semester : Valid